The commercial offerings mentioned in this section are widely used, mostly by the business side of the house, rather than the engineering team. However, in the context of observability, we do need a comprehensive approach, enabling as many as possible job families to participate and benefit from the system signals.

6.2 All-in-ones

Technically speaking, all the cloud provider offerings, including CloudWatch, X-Ray, Azure Monitor, and Google Cloud's operations suite, fall into the all-in-ones category. We covered them in chapter 5, so let's focus here on the other offerings in this space.

There are some open source offerings, with some of them hosted by the Cloud Native Computing Foundation (CNCF) or the Apache Software Foundation (ASF), that offer support for two or more signal types. They come with ingestion and query capabilities, typically allowing different backends to be plugged in or in-memory (nonpersistent) storage.

6.2.1 CNCF Jaeger

The CNCF Jaeger (https://github.com/jaegertracing/jaeger) project started out at Uber and was inspired by Google's Dapper project and Zipkin. Nowadays, it is a graduated CNCF project that positions itself as a distributed tracing platform, with built-in support for various backends, including Apache Cassandra; ES/OS; and custom ones, like ClickHouse, via the third-party storage plug-in framework.

Jaeger's overall architecture is a collection of microservices, written in Go, that are working together. The core Jaeger components are

- *Agent (optional)*—A daemon sends traces (over UDP) to the collector, designed to be deployed to all hosts where apps run. The agent is being replaced by OpenTelemetry.
- *Collector*—Receives traces from OpenTelemetry (SDKs or collector) or Jaeger agents and validates, enriches, and stores them in a storage backend.
- *Query service*—Exposes the APIs for retrieving traces from storage and hosts the UI.
- *UI*—The web-based GUI a user interacts with to search for and analyze traces and/or view the system architecture (graph representation of the microservices).
- *Ingester*—A simplified version of the collector that reads traces from Apache Kafka (a popular open source message broker) and writes them to a storage backend.

For experimentations (dev–test use cases), Jaeger also supports pure in-memory storage, readily usable via an the "all-in-one" distribution, combining the core Jaeger components in one package.